

Topic 9 Summary: Moments and Couples/ static equilibrium

- What is static equilibrium?
- Definitions
 - Rigid body
 - Moment of a force
- The principle of moments
- Conditions for static equilibrium
- Couples:
 - Definition
 - Property of couples and proof
 - Moment of a couple
 - Combination of couples
 - Two theorems on couples
- Resultant of systems of parallel forces
- Worked examples

Systems of coplanar forces

A system of coplanar forces has 3 options:

It either reduces to:

- a) a single resultant force,
- b) a resultant couple, **OR**
- c) the system is in equilibrium.

Conditions for static equilibrium

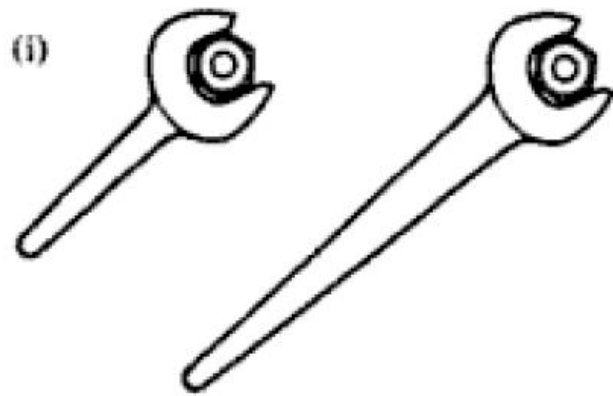
No linear acceleration:

$$\sum \vec{F} = 0 \Rightarrow \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned}$$

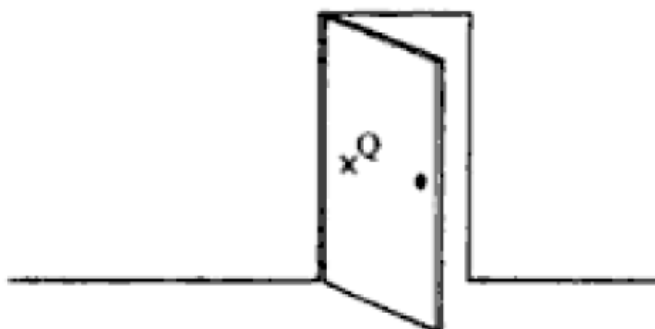
No angular acceleration:

$$\sum \vec{M}_z = 0$$

This is for two dimensional problems where all forces act in one plane, i.e. the xy-plane.



(iii)



Equilibrium of particles and rigid bodies under coplanar forces

Def: Rigid body

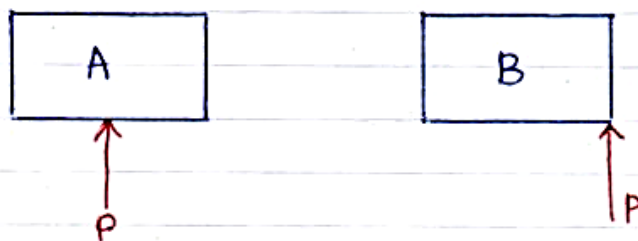
In physics, a rigid body is an idealization of a solid body in which deformation is neglected. In other words, the distance between ^{any} two given points of a rigid body remains constant in time regardless of external forces exerted on it.

When a particle of a rigid body is at rest under a set of coplanar forces, then it is said to be in static equilibrium. In this state, the vector sum of the forces on the particle or rigid body is zero.

Moment or Turning Effect of a Force.

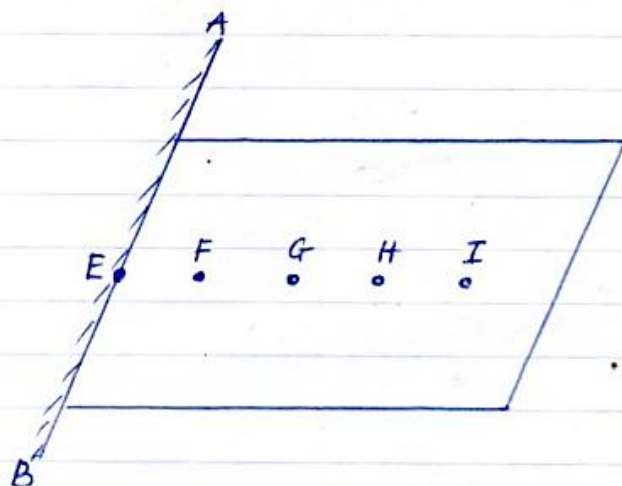
When analysing the motion of a body, the position of the point of application of a force must be considered as well as the magnitude and direction of the force.

Consider the fol:



If equal forces are applied at the positions shown on equal objects A and B, A will slide in a straight line (translation) while B will not slide in a straight line (translate + rotate).

Consider the diagram below representing a horizontal trap door with AB as the line of hinge.



The trap door may be opened by means of an upward pull P applied at rings attached at points F, G, H or I. Suppose $EF = FG = GH = HI = 0.25 \text{ m}$ and the trap door is uniform and weighs 120 N . If the hinge is well greased, the following experimental results are obtained.

Position of force	F	G	H	I
Distance of hinge from force (x metres)	0.25	0.50	0.75	1.00
Size of force (P Newtons)	240	120	80	60

The force P gets less as the distance x increases. However the product Px is ^{the same} throughout. This product is called the **moment** or **turning effect** of the force P about the hinge. (Units: Nm).

Def: Moment of Force.

The moment of a force about a point is the magnitude of the force multiplied by the perpendicular distance of the point from the line of action of the force.

When the forces are in the plane of the paper, the directions of their moments are given either as clockwise (negative) or anticlockwise (tre).

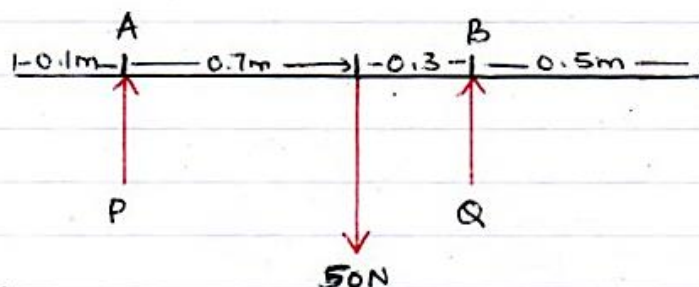
The algebraic sum of the moments of a system of coplanar forces about any point is equal to the moment of their resultant about that point.

The principle of moments states:

"For a body in equilibrium under a system of forces, the sum of the moments of the forces about any axis, is zero."

Example 1:

A uniform rod of mass 5 kg and length 1.6 m is laid horizontally across smooth supports at A and B. A is 10 cm from one end and B is 50 cm from the other end. Find the forces at A and B.



The forces at A and B are vertical. Let the weight of the rod act at the centre of the rod (50 N) Since the rod

is uniform.

Since the rod is in equilibrium, the sum of the forces is zero.

$$\text{i.e. } P + Q - 50 = 0 \quad \dots (i)$$

also

Since the rod is in eqbm, the sum of the moments of the forces about any axis is zero.

Summing moments about A,

$$Q \times (1.0) - 50(0.7) = 0$$

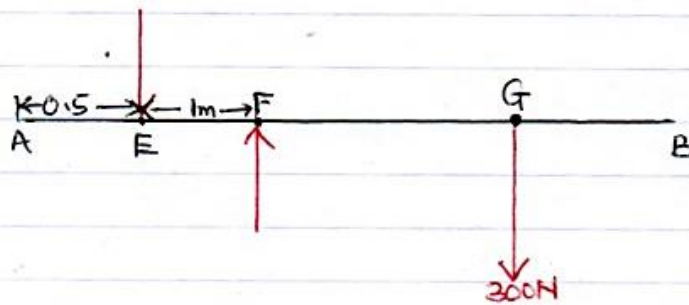
$$Q - 35 = 0$$

$$\therefore Q = \underline{35\text{ N}}$$

$$\text{from eqn(i), } P = \underline{15\text{ N}}$$

i.e. the contact forces at A and B are $\approx 35\text{ N}$ & 15 N .

2. A non-uniform plank AB of mass 30 kg and length 4 m has its centre of gravity at G. The plank is kept horizontal by parallel bars at E and F, 1 m apart so that $AE = 0.5\text{ m}$.



If the forces of the bars on the plank are vertical and act as shown, and the ratio of the sizes of these forces is $5:3$, find:

- the forces on the bars
- the position of G.

Sol:

NB: The larger of the two forces must be at F as it is opposed by two other forces.

- a) Suppose the larger force is $5P$ N. Then the smaller one is $3P$ N. As rod is in eqbm, sum of forces is zero.
i.e. $5P - 3P - 300 = 0$
 $\Rightarrow P = 150$ N.

$\therefore$ force on bar at E = $3 \times 150 = 450 \text{ N}$.
and " " F = $5 \times 150 = 750 \text{ N}$.

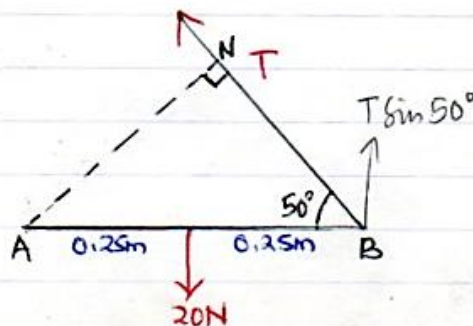
- b) Let $AG = x \text{ m}$, then $FG = (x - 1.5) \text{ m}$

$$\begin{aligned}\text{Moments about F: } 450 \times 1 - 300(x - 1.5) &= 0 \\ 450 - 300x + 450 &= 0 \\ 300x &= 900 \\ \underline{x} &= 3 \text{ m}\end{aligned}$$

i.e. the centre of gravity G is 3m from the end A .

Example 3:

A rod of length 50 cm is smoothly hinged at one end A and is kept in a horizontal position by a string attached to the other end B, so that the string is inclined at 50° to BA. If the rod is uniform and weighs 20 N, find the tension in the string.



If the tension is T N, its moment about A will be $T \times AN$ where AN is $\perp$ to the string

$$AN = 0.5 \sin 50 = 0.383 \text{ m.}$$

Taking moments about A,

$$T \times 0.383 - 20 \times 0.25 = 0$$

$$0.383T = 5$$

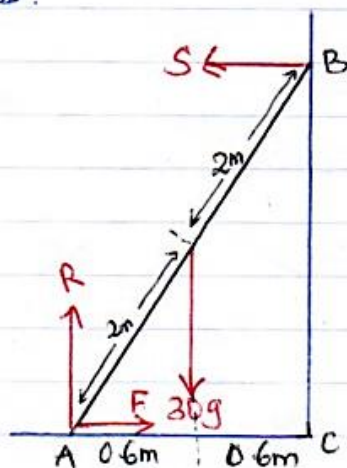
$$T = \underline{\underline{13.05 \text{ N}}}$$

Example 4:

A uniform ladder of mass 30 kg and length 4 m stands on a rough horizontal ground and leans against a smooth vertical wall. The foot of the ladder is 1.2 m away from the wall. Find the normal contact force of the wall on the ladder and of the ground on the ladder. Find also the frictional force of the ground on the ladder.

Sol:

FBD.



R & S - Normal contact forces at A & B respec
 F - Frictional force at A.

Since the ladder is in eqbm, sum of forces (resultant) in both ~~the~~ directions is zero.

Taking components.

$$\begin{pmatrix} F \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ R \end{pmatrix} + \begin{pmatrix} -S \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -30g \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Equating components:

$$F - S = 0 \Rightarrow F = S$$

$$R - 30g = 0 \Rightarrow R = 30g = \underline{\underline{300\text{N}}}.$$

By Pythagoras theorem on $\triangle ABC$,

$$BC^2 = 4^2 - 1.2^2 \therefore BC = 3.8158\text{ m}.$$

Taking moments about A,

$$S \times 3.82 - 300 \times 0.6 = 0$$

$$S = \frac{300 \times 0.6}{3.82} = \underline{\underline{47.17\text{N}}}.$$

$$F = S \Rightarrow F = \underline{\underline{47.17\text{N}}}$$

Example 5:

A uniform ladder of weight W rests at an inclination to the horizontal of θ , where $\tan \theta = \frac{5}{4}$, with its foot on rough horizontal ground and its upper end against a rough wall. μ for both ground and wall is $\frac{1}{2}$. How far up the ladder can a man of weight W climb before it slips. **Ans ($\frac{9}{10}$ up of the ladder).**

Sol:

Let the length of the ladder be $2a$ and the man be a distance x from its foot when it is on the verge of slipping.

COUPLES

A couple is a system consisting of two equal and opposite forces, with a different line of action.

E.g. forces on a propeller blade.

Property:

A couple has the same moment about every point.

Proof:

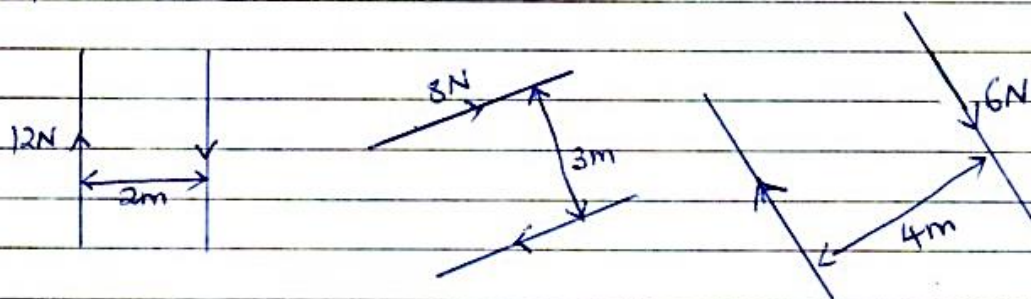
Consider the diagram below



The moment of the couple shown about any point A is $F(d+x) - F(x) = Fd$ anticlockwise

i.e. moment of a couple = one of its forces multiplied by the distance apart of the forces.

A couple in a given plane is completely defined by its moments i.e. there is no need to state the value, actual size, direction or distance apart of its two forces. E.g. the following three couples are equivalent each being describable as a clockwise couple of 24 Nm .



A couple has no resultant force in any direction. It is a basic unit in mechanics and cannot be simplified further. If a system of forces is equivalent to a couple, that couple is the resultant.

Combination of Couples.

Provided that they are coplanar, the moments of couples can be added or subtracted to obtain resultant couples.

Theorems

1. Any force, together with a coplanar couple is equivalent to an equal parallel force.
2. Any force is equivalent to an equal parallel force together with a couple.

Resultant of Systems of Parallel Forces.

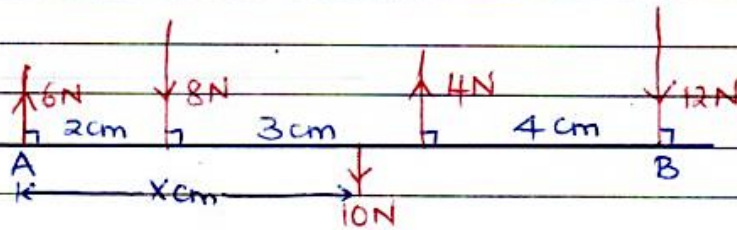
To specify any resultant force completely, its magnitude, direction, and line of action must be stated. For parallel forces, the direction is that of the original forces and the magnitude is the algebraic sum.

To find the line of action, we use the theorems on moments which may be simplified to the following important principle:

moment of resultant = moment of original system
(about any point)

Example 1:

Find the line of action of the resultant of the system shown below if the resultant is a force of 10N in the downward direction.



Sol

Let the resultant cut AB at a distance x to the right of A.

Using the principle

Moment of resultant = moment of original system.

Taking moments about A,

$$-10x = -(8)(2) + (4)(5) - (12)(9)$$

$$-10x = -104$$

$$x = 10.4 \text{ cm.}$$

Q: If an anticlockwise couple of 20Nm is added to the system above, find the resultant.

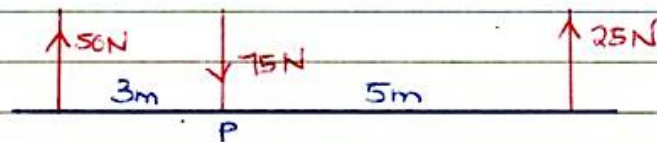
$$-10x = -104 + 20$$

$$-10x = -84 \Rightarrow x = 8.4 \text{ cm.}$$

i.e. the resultant is a downward force of 10N at a point 8.4 cm from A.

Example 2

Show that the resultant of the system shown below is a couple and find its moment.



Sol:

Since the algebraic sum of the forces is zero ($50 - 75 + 25 = 0$), the system does not ~~have~~ ^{reduce to} a resultant force.

It must $\therefore$ reduce to a couple or be in equilibrium.

i.e. If the moments about a point are not zero, then the system is not in eqbm, and must be equivalent to a couple of the moment found.

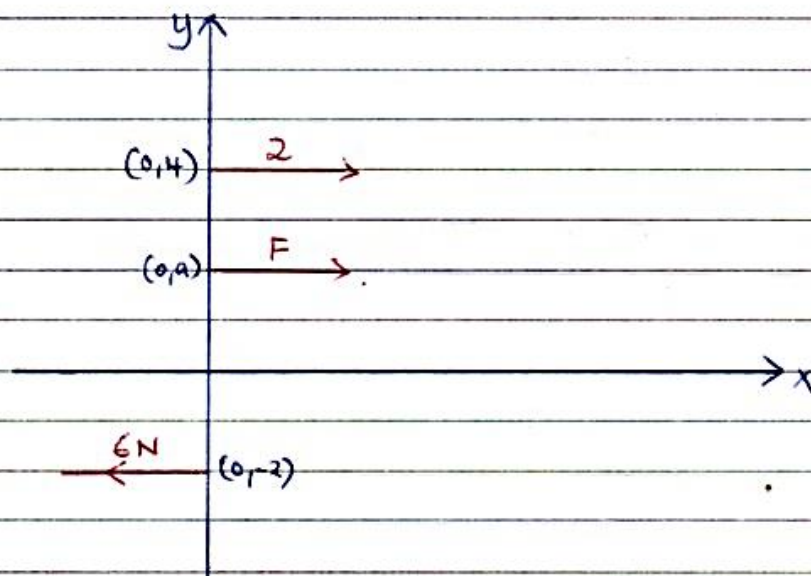
Moments about P.

$$(25)(5) - (50)(3) = -25 \text{ Nm.}$$

$\therefore$ Resultant is a clockwise couple of moment 25 Nm.

Example 3

Forces $-6\hat{i}$, $2\hat{i}$ and $F\hat{i}$ act at the points $(0, -2)$, $(0, 4)$ and $(0, a)$ as shown below. Given the resultant is a clockwise couple of moment 30, find P and a .



Sol:

Since the sum is equivalent to a couple, the sum of forces in any direction is zero.

$$\therefore 2 + F = 6 \Rightarrow \underline{F = 4 \text{ N}}$$

Taking moments about origin (clockwise).

$$(6)(2) + (F)(a) + (2)(4) = 30$$

$$12 + 4a + 8 = 30$$

$$4a = 10$$

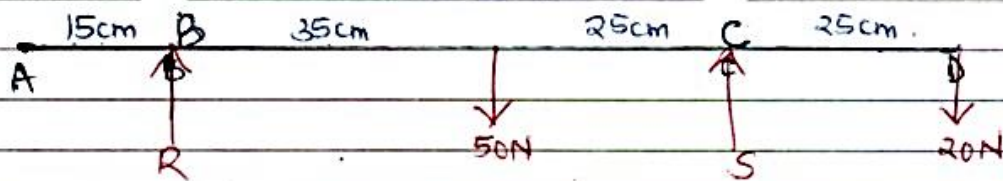
$$a = \underline{\underline{2.5}}$$

Example 4

A uniform rod AD of length 1m and mass 5kg rests horizontally on supports B and C where AB = 15cm and CD = 25cm. The rod carries a ^{body of} mass 2kg at D. Find:

- The forces on the supports
- The couple which will just make the rod tip about C and the new values of the forces on the support when it is acting.

Sol:



a). Taking moments about c

$$50(25) - R(60) - 20(25) = 0 \dots$$

$$R = \underline{12.5 \text{ N.}}$$

Since there is no resultant force, $R + S = 70$

$$\Rightarrow S = \underline{57.5 \text{ N.}}$$

b). Let the req couple have an anticlockwise moment of L .

If the rod is about to tip about c, the reaction at B must be zero.

Taking moments about c,

$$L + 50(25) - 20(25) = 0$$

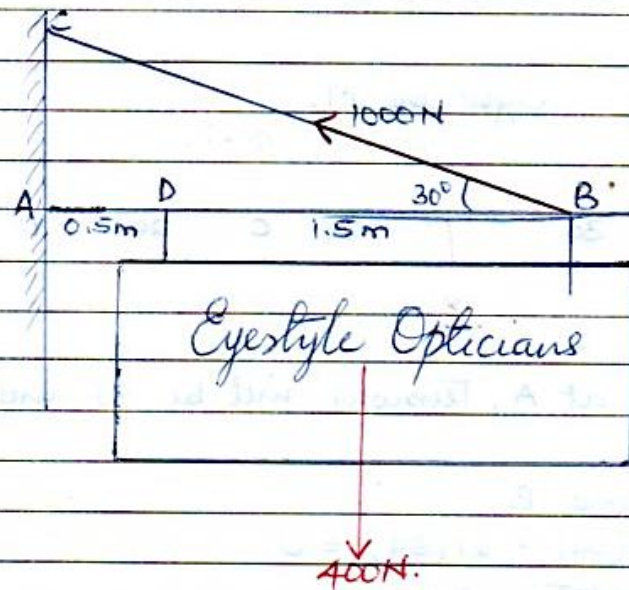
$$L = -750 \text{ Nm.}$$

$\therefore$ Required couple is a clockwise couple of 750 Nm.

Since the couple provides no extra vertical force, the total downward force of 70N has still to be balanced. The forces on the supports must be zero and 70N.

Example 5:

A shop sign is supported from a light pole AB embedded in a wall at A with a chain CB as shown. The sign weighs 40 kg and the tension in the string is 1000 N. Using the dimensions shown, find the couple which will be needed to maintain equilibrium ($g = 10 \text{ m/s}^2$).



Sol:

Let couple have moment M Nm. Assume sign to be uniform & supported symmetrically. So that the weight will be divided equally between B and D.

Taking moments about A.

$$M + 1000(2\sin 30) - 200(0.5) - 200(2) = 0$$

$$M + 1000 - 100 - 400 = 0$$

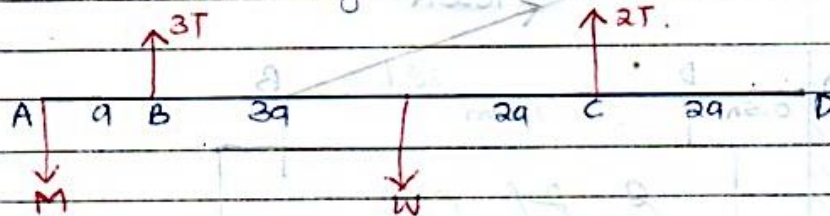
$$M = 500 \text{ Nm}$$

The couple is approx 500 Nm.

Example 6:

A uniform beam AD of length $8a$ and weight W is suspended horizontally by 2 vertical strings at B and C where $AB = a$ and $CD = 2a$. When a certain weight is placed at A, the tensions are in the ratio $3:2$. Where must the same weight be placed to equate the tensions?

Sol: Let unknown weight be M .



If the weight M is at A, Tensions will be $3T$ and $2T$.

Taking moments about B,

$$M(a) - W(3a) + 2T(5a) = 0$$

$$Ma + 10Ta = 3Wa$$

$$M + 10T = 3W \quad \text{--- (i)}$$

Since there is no resultant vertical force,

$$5T = W + M \quad \text{--- (ii)}$$

Eliminating T from (i) & (ii)

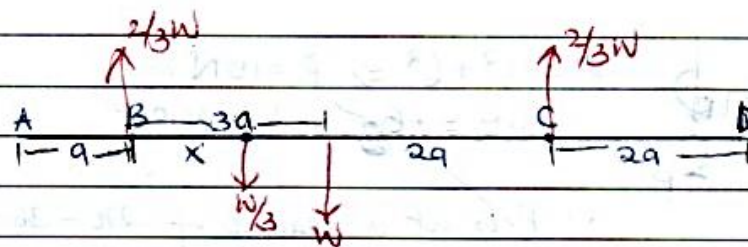
$$M + 10 \left(\frac{W+M}{5} \right) = 3W$$

$$\boxed{M = \frac{W}{3}}$$

Wherever the weight $W/3$ is placed, total downward force is $4/3W$.

If the tensions are equal, they must each be $2W/3$.

Let the weight be a distance x from to the right of B,



Taking moments about B,

$$(2/3W)(5a) - (W/3)(x) - W(3a) = 0$$

$$\frac{10Wa}{3} - \frac{Wx}{3} - 3Wa = 0$$

$$10/3a - x/3 - 3a = 0$$

$$\Rightarrow a/3 = x/3 \Rightarrow x = a$$

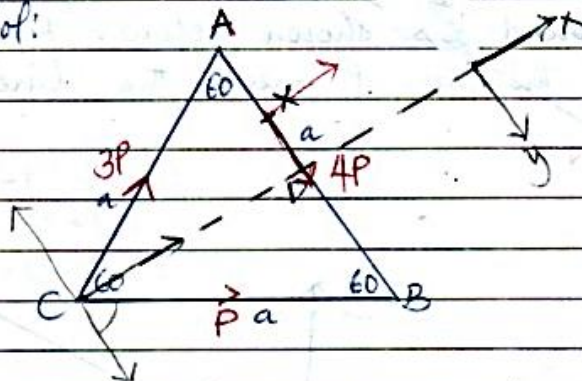
The weight should be placed a units from to the right of B.

Example 7:

Forces of $4P$, P and $3P$ act along the sides AB , CB and CA of an equilateral triangle ABC of side a .

Find the magnitude of the resultant, the angle it makes with AB , and the point at which its line of action meets AB .

Sol:

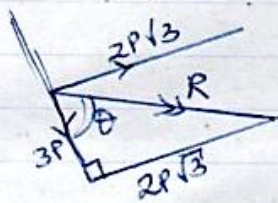


Total components in dir AB

$$4P + P \cos 60 - 3P \cos 60 = 3P$$

Total components in dir CD

$$3P \sin 60 + P \sin 60 = 4P \sin 60 = \underline{\underline{2P\sqrt{3}}}$$



$$R^2 = 9P^2 + 4P^2(3)$$

$$R = 4.5825P$$

$$\tan \theta = \frac{2P\sqrt{3}}{3P} \Rightarrow \theta = \underline{\underline{49.11^\circ}}$$

Let the line of action cut AB at a distance x from A.

Moment of resultant = moment of original system.

Taking moments about A.

$$2P\sqrt{3}(x) = P(a \sin 60^\circ)$$

$$2P\sqrt{3}x = Pa \frac{\sqrt{3}}{2}$$

$$\therefore x = \underline{\underline{a/4}}$$

Ex:

Forces of 36N, 15N, 24N, 10N act along the sides AB, CB, DC, DA ^{respect.} of a rectangle ABCD in which AB = 12cm & BC = 5cm. A force of 65N acts along the diagonal BD. Prove that the system is in eqbm.